

Lecture 3: Limit of Sequence

Mathematical Analysis

1 Convergent Sequences

Definition 1.1 (Convergent Sequence). [def:convergent-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$, where F is a ordered field, is said to be *convergent* if

$$\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists n \in \mathbb{N}_1, \forall m \in \mathbb{N}_1, m \geq n, a_m \in V_\varepsilon(\ell).$$

Remark 1.1. The intuition of convergence is that we can find a value ℓ such that we can propose any $\varepsilon > 0$, such that there exists an index $n \in \mathbb{N}_1$, for all $m \in \mathbb{N}_1$ where $m > n$,

Remark 1.2 (Convergence Over a General Ordered Field). [rem:convergence-ordered-field] Results in this lecture are stated for a general ordered field F by default; theorems that require a complete ordered field will say so explicitly in their hypotheses. Recall that $\mathbb{Q}$ is an ordered field and $\mathbb{R}$ is a complete ordered field.

2 Limit of a Sequence

3 Bounded Sequences

4 Monotone Convergence Theorem